JAIN Knowledge Campus, Jayanagar 9th Block, Bengaluru

DEPARTMENT OF COMPUTER SCIENCE AND IT

MCA PROGRAMME

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EXP: 9 - EVALUATE A FULL DATA PIPELINE FROM RAW DATA INGESTION TO VISUALIZATION IN POWER BI

NAME: ABHISHEK KUMAR USN: 25MCAR0141

SEM: III SPEC: SCT

SUBJECT: CLOUD ANALYTICS-LAB

CODE: 25MCASC302L

FACULTY NAME: Dr. PRABHAKARAN ABRAHAM, ASSOCIATE PROFESSOR

1 Aim

To evaluate a full data pipeline from raw data ingestion to visualization in Power BI, using the student performance dataset already used in Experiments 07 and 08, so that every stage can be checked against known values instead of against invented data.

2 Objective and Theory

The objectives are to ingest a raw comma-separated values file from Azure Blob Storage through Azure Data Factory (ADF) into Azure SQL Database, validate the row count at each stage, connect Power BI to the resulting SQL table, build dashboard visuals, and evaluate whether the pipeline preserves the data end to end.

Azure Data Factory (ADF) is a cloud data integration service for building pipelines that move and transform data [1]. Azure SQL Database is a managed relational database service in Microsoft Azure, and it supports standard Transact-SQL (T-SQL) queries, tables, and aggregations [2]. Power BI is a business analytics service for creating interactive reports and dashboards. In this experiment, the pipeline is Blob Storage to ADF to Azure SQL Database to Power BI, with a validation count at each stage: 50 rows in the source file, 50 rows read and copied by ADF, 50 rows in the SQL table, and 50 students in the Power BI report. The connection mode actually used in Power BI was Import, verified in the dataset settings, so updated data reaches the report through a semantic model refresh, not through a live query at visual load time.

The dataset contains 50 student records with scores in mathematics, reading, and writing, attendance rate, study hours per week, and a final grade from A to F. Two fields have missing values in the source file, attendance rate and study hours per week, each with 45 non-null values out of 50. Those missing values are kept as null values, and averages are calculated over the non-null values, which is stated here so that the averages in Section 5 can be reproduced exactly.

3 Architecture Flow

Data flows from Azure Blob Storage through ADF into Azure SQL Database, and Power BI reads the SQL table through a semantic model refresh, as shown in Figure 1.1. Table 1.1 lists the resources used.

Table 1.1 Resources used in Experiment 09

Component	Resource name	Role
Resource Group	25MCAR0141	Logical container for all experiment resources.

Component	Resource name	Role
Storage Account	calablake141	Azure Data Lake Storage Gen2, rawdata container, student_performance.csv.
Data Factory	calab-adf-141	Pipeline PL_Exp09_FullPipeline, Copy activity.
SQL Server	calab-sql-141.database.windows.net	Azure SQL Server, Central India, version 12.0.
SQL Database	calab-db-141	Basic, 5 DTU, table StudentPerformance_Exp09.
Power BI	Power BI Service, Import	Report Exp09_PowerBI_FullPipeline, five visuals, semantic model refresh.

Source: Azure resource list, ADF run record, and Power BI report record, verified 2026-11-02.

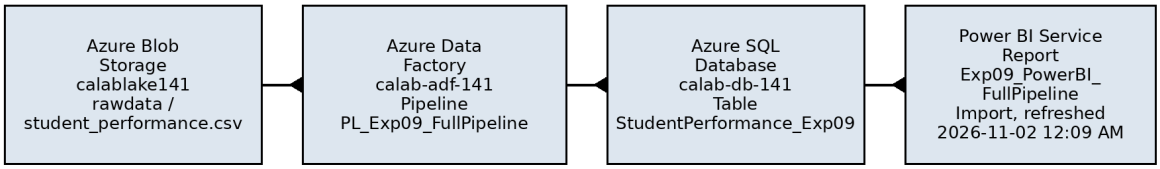


Figure 1.1 Experiment 09 data flow

Blob Storage to ADF to Azure SQL Database to Power BI, with semantic model refresh.

4 Procedure

Step 1 (ADF datasets). In Data Factory calab-adf-141, create dataset DS_Exp09_Blob_CSV for rawdata/student_performance.csv, using linked service LS_Exp07_Blob, with first row as header, and dataset DS_Exp09_Sql_Table for table StudentPerformance_Exp09, using linked service LS_Exp07_Sql.

Step 2 (ADF pipeline). Create pipeline PL_Exp09_FullPipeline with a Copy activity, CopyBlobToSqlExp09, with DS_Exp09_Blob_CSV as source and DS_Exp09_Sql_Table as sink. Add a preCopyScript that creates table StudentPerformance_Exp09 if it does not exist, with the columns listed in create_table_exp09.sql in the GitHub folder. A new table name is used, instead of reusing StudentPerformance from Experiment 07, so that the row count for this experiment is exactly 50, not duplicated by a second load.

Step 3 (ADF run and validation). Run the pipeline. The run succeeded, as recorded in Table 1.2, with rowsRead 50 and rowsCopied 50. This validates the ingestion stage.

Step 4 (Power BI connection). In Power BI Service, create a report from Azure SQL Database with server calab-sql-141.database.windows.net and database calab-db-141, using Microsoft account authentication, and select table StudentPerformance_Exp09. The dataset settings show SqlServer calab-sql-141.database.windows.net, database calab-db-141, with no gateway needed, because all data sources are in the cloud. Last refresh succeeded on 2026-11-02 at 12:09:38 AM, and the Refresh section states that a refresh schedule imports data from the data source into the semantic model, which is the Import mode evidence.

Step 5 (Dashboard visuals). Build five visuals from table StudentPerformance_Exp09: KPI cards for total students and average scores; a clustered column chart of average mathematics, reading, and writing scores by FinalGrade; a bar chart of student count by FinalGrade; a donut chart of student share by Gender; and a table of FinalGrade, student count, and average scores. The queries for these visuals are in dashboard_queries.sql in the GitHub folder.

Step 6 (End-to-end evaluation). Compare the counts at each stage: source file 50 rows, ADF rowsRead 50, ADF rowsCopied 50, Power BI student count 50, grade counts sum to 50, gender counts sum to 50, and table totals match the KPI cards. The pipeline therefore preserves the data end to end, with no rows lost and no rows duplicated, within the limits of the 50-row test file.

5 Proof of Work

The GitHub repository for this experiment is https://github.com/25MCAR0141/Cloud-Analytics-Azure/tree/main/Lab/Exp_09 . The repository contains the README, architecture diagram, Power BI screenshots, ADF pipeline definition, SQL scripts, and dashboard queries, without connection secrets. Table 1.2 records the ADF execution, and Table 1.3 records the dashboard data calculated from the 50 rows in StudentPerformance_Exp09.

Table 1.2 ADF pipeline execution, Experiment 09

Item	Value
Pipeline	PL_Exp09_FullPipeline
Successful run	b2b9f84b-31b9-49d6-b6b5-6c0ab25e477a, Succeeded, 2026-11-02, duration 24,707 ms

Item	Value
Copy activity	CopyBlobToSqlExp09, Succeeded, activity run 28e763d5-d60d-43b0-a56b-9a26d828cc45
Rows read / copied	50 / 50
Data read / written	3,174 bytes / 4,754 bytes
Copy duration	11 seconds, 4 Data Integration Units, AutoResolve Central India

Source: ADF pipeline run and activity run records, 2026-11-02.

Table 1.3 Dashboard data, average scores by FinalGrade

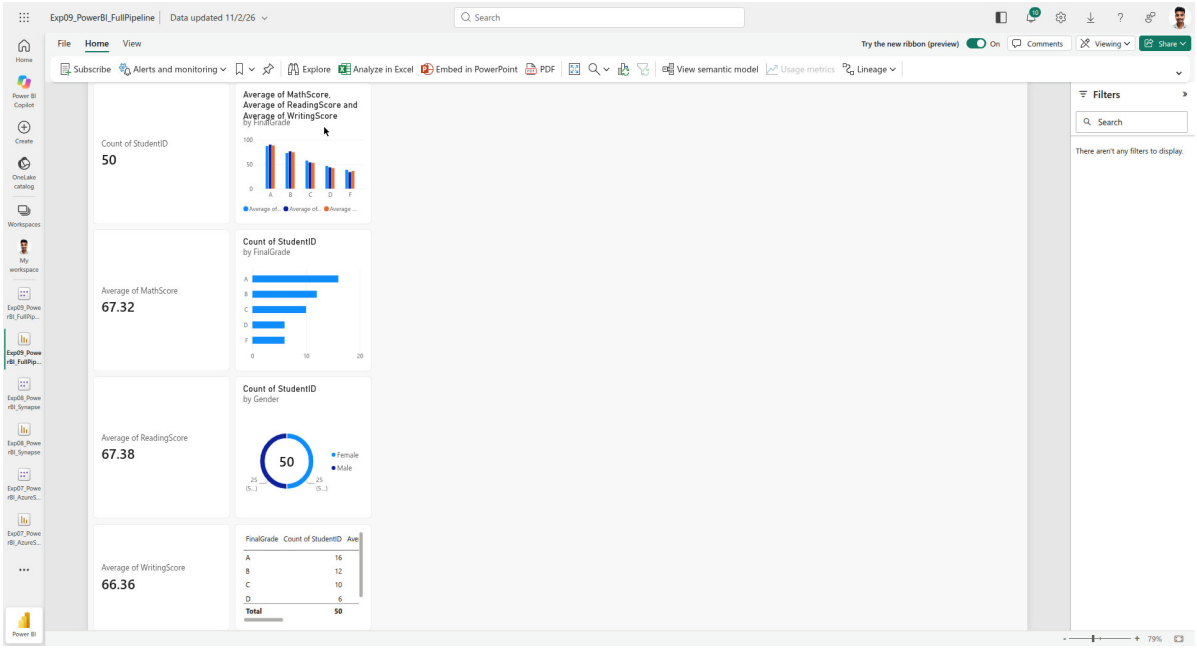
FinalGrade	Students	Avg MathScore	Avg ReadingScore	Avg WritingScore
A	16	87.50	90.38	88.63
B	12	73.17	76.58	74.83
C	10	57.60	53.80	53.10
D	6	46.67	43.67	42.17
F	6	38.67	34.00	36.33

Source: calab-db-141, table StudentPerformance_Exp09, 50 rows, verified 2026-11-02. Overall averages are MathScore 67.32, ReadingScore 67.38, and WritingScore 66.36.

Power BI report record. The saved Power BI Service report is Exp09_PowerBI_FullPipeline, at https://app.powerbi.com/groups/cfa9da9e-3370-4b81-b46c-b14de945b284/reports/d5f5e8a6-ad9b-4ccf-8b52-19415661c3ef/c019345000591b7a516d?experience=power-bi , with semantic model Exp09_PowerBI_FullPipeline, id 2c19698a-c0b2-4818-b9d7-38e7b5bc6f38, refreshed 2026-11-02 at 12:09:38 AM. Page 1 contains five verified visuals: KPI cards for student count 50 and average scores 67.32, 67.38, and 66.36; a clustered column chart of average scores by FinalGrade; a clustered bar chart of student count by FinalGrade; a donut chart of student share by Gender, Female 25 and Male 25; and a table of FinalGrade, student count, and average scores, with total 50, 67.32, 67.38, and 66.36.

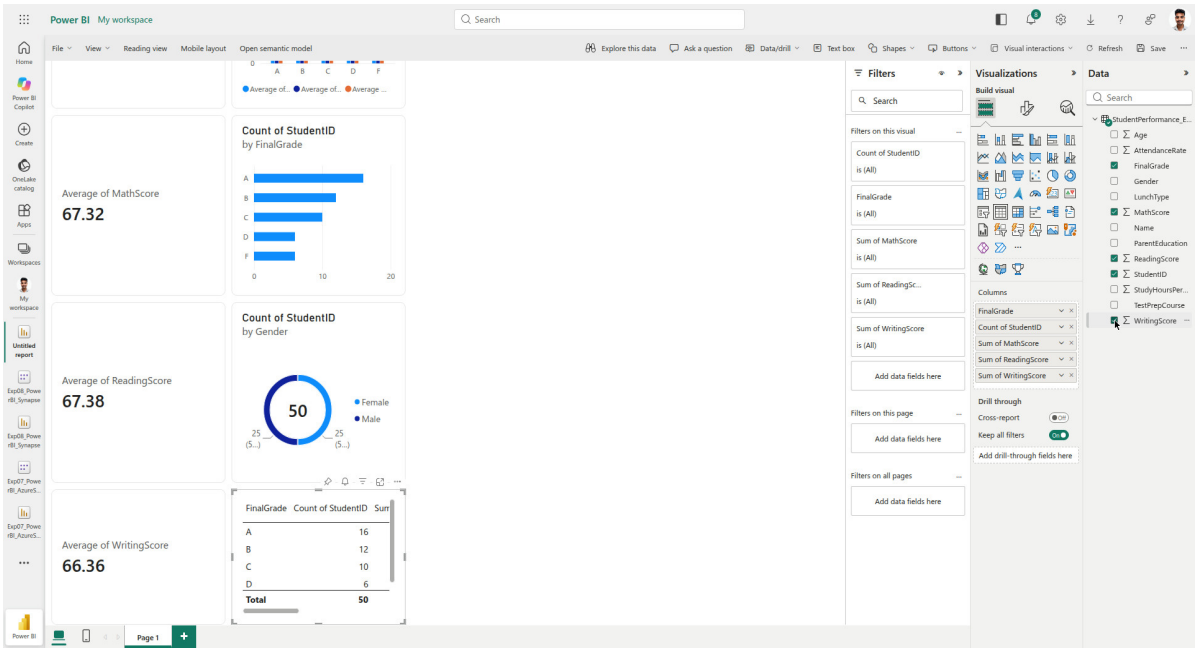
6 Screenshots

The screenshots in this section are real browser screenshots of the saved Power BI Service report, captured from the signed-in Power BI session on 2026-11-02, not replotted charts and not cropped chart images. The report title Exp09_PowerBI_FullPipeline is visible in the report screenshot, and the report URL is given in Section 5. The browser capture shows the Power BI viewport, not the operating system browser chrome, so the address bar and tabs are not drawn inside the images, as in Experiments 07 and 08. Figure 1.1 in Section 3 remains the architecture diagram, and Tables 1.2 and 1.3 in Section 5 remain the execution and data records; this section is the visual proof of the dashboard itself.



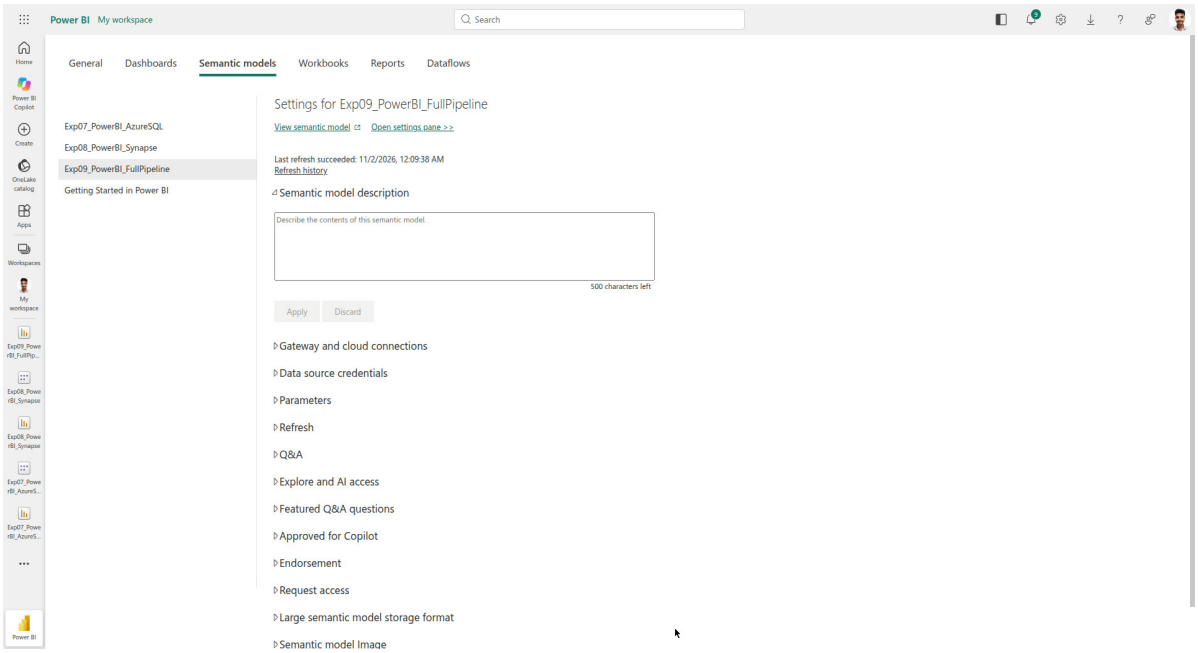
Screenshot 1 Power BI Service report Exp09_PowerBI_FullPipeline, Page 1, reading view

All five visual groups in one view: KPI cards for student count 50 and average scores 67.32, 67.38, and 66.36, clustered column chart by FinalGrade, bar chart by FinalGrade, donut chart by Gender, and FinalGrade table with total 50.



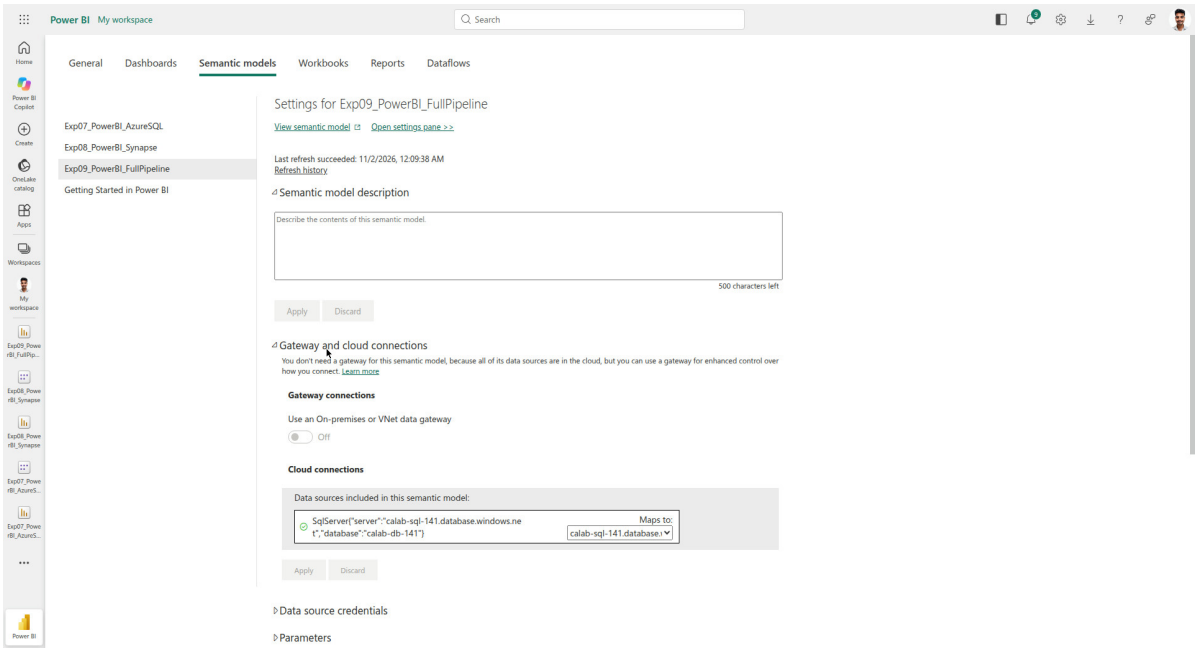
Screenshot 2 Power BI Service report Exp09_PowerBI_FullPipeline, Page 1, editing view, lower part

KPI cards for average scores, bar chart of student count by FinalGrade, donut chart by Gender with Female 25 and Male 25, and the FinalGrade table.



Screenshot 3 Power BI semantic model settings for Exp09_PowerBI_FullPipeline

Settings page showing Last refresh succeeded 11/2/2026, 12:09:38 AM.



Screenshot 4 Power BI semantic model cloud connection for Exp09_PowerBI_FullPipeline

Cloud connection showing SqlServer calab-sql-141.database.windows.net, database calab-db-141, with no gateway needed, because all data sources are in the cloud.

Cost note. No new billable resource was created for this experiment. ADF used 4 Data Integration Units for 11 seconds, Azure SQL Database calab-db-141 is Basic, 5 DTU, and Power BI Service is the existing student workspace. Keep the SQL database only until submission is complete, then delete it and, if no other experiment needs it, the server, to stop the hourly charge.

References

[1] Microsoft, “Azure Data Factory documentation,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/data-factory/

[2] Microsoft, “Azure SQL Database documentation,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/azure-sql/database/

[3] Microsoft, “Power BI documentation,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/power-bi/